

## Supplementary Information (integration draft)

**Status:** portfolio-level disposition is imported from NeuroMech Route-D evidence matrix V3 (39 records). This supplement includes the complete 39-row portfolio disposition and V1 run records, but it is not yet a fully adjudicated source-to-claim supplement. Selected records from the divergent experiment-publication branch are listed separately in `../evidence/PUBLICATION_BRANCH_CROSSWALK.csv`. C04–C06 have selected-file hash pins but no exact V3 row. C08's pinned contract identifies it as a post-result follow-up informed by, and distinct from, M5\_ELIGIBILITY\_TEMPORAL\_XOR\_V2; its relationship to the M10 record remains unresolved. Individual selected-file hashes and a passing stored-output verification are reported, while the package-root checksum list retains stale/unavailable entries. The C08 evidence identity is therefore narrowed but the full source-package audit is not closed.

### Supplementary Table S1 — Full Route-D portfolio disposition

`../supplementary/TABLE_S1_PORTFOLIO_DISPOSITION.csv` contains all 39 rows from `PROJECT_EVIDENCE_CLASSIFICATION_MATRIX_V3.csv`, with an added proposed NeuroConverge disposition. It preserves the original labels, portfolio scope, units, source pointers and interpretation limits. The imported source matrix hash and source worktree state are recorded in `../workstreams/PAPER_CRITICAL_SOURCE_SNAPSHOT.json`.

The portfolio includes the bounded M0 assay; Fish1.5 primary, invalid-null, robustness and post-result records; published worm evidence and schema-limited analyses; M2 synthetic rounds and historical failure; M4–M11 algorithmic studies; M12 task-viability failure; Drosophila-inspired synthetic tests; and separately governed RR18/RR19 records. Different studies are not replications and are not pooled. RR18/RR19 remain separate workstreams.

### Supplementary Table S2 — Publication-branch crosswalk

`../evidence/PUBLICATION_BRANCH_CROSSWALK.csv` maps the nine claims from the initial NeuroConverge draft to exact or unresolved Route-D matches. The disposition is: C01–C03 map to V3 main-text candidates; C04–C06 are identified branch packages retained in the Supplementary Information without exact V3 rows; C07 is an editorial synthesis, not a source record; C08 is a post-result follow-up whose relationship to M10 remains unresolved; and C09 maps to a V3 supplementary-only record. “No exact row” means there is no exact counterpart in the 39-row V3 matrix. It does not imply that selected files are unidentified. No branch-only package is merged into V3 disposition counts. RR18/RR19 are present as separately governed entries in the V3 disposition inventory but excluded from this synthesis; their appearance in that inventory is not an inclusion as evidence for the manuscript.

| Initial-draft claim group | Count | Manuscript disposition                                           | Selection rationale / unresolved boundary                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|-------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C01–C03                   | 3     | Main-text evidence candidates                                    | Exact V3 matches for the bounded connectome assay, Fish1.5 records and published worm evidence/schema audit; distinct evidence types remain separated.                                                                                                                                                                                                                                                                                                 |
| C04–C06                   | 3     | Supplementary branch records                                     | Selected packages are identified and hash-pinned, but have no exact row in V3; retained for transparency without merging into the 39-row counts.                                                                                                                                                                                                                                                                                                       |
| C07                       | 1     | Organizing synthesis in main text                                | A framing proposal, not an empirical record or validated hierarchy; requires novelty assessment.                                                                                                                                                                                                                                                                                                                                                       |
| C08                       | 1     | Supplementary post-result follow-up; M10 relationship unresolved | Its pinned contract describes a distinct 32-seed/five-arm follow-up informed by the earlier 30-seed/three-arm M5_ELIGIBILITY_TEMPORAL_XOR_V2 ( <a href="#">C08 contract</a> ; <a href="#">V2 summary</a> ). The pinned publication branch does not provide the M10 final-status artifact needed to resolve that relationship. Selected-file hashes and stored-output verification do not cure stale/unavailable entries in the C08 root checksum list. |
| C09                       | 1     | Supplementary V3 record                                          | Exact V3 record exists but is classified supplementary-only in the source matrix.                                                                                                                                                                                                                                                                                                                                                                      |
| RR18/RR19                 | —     | Separately governed exclusions                                   | They are recorded in the inventory where applicable, but excluded from this manuscript synthesis; no ownership boundary is changed here.                                                                                                                                                                                                                                                                                                               |

This accounting closes the disposition of the nine initial-draft claim groups for reporting purposes. It also resolves one earlier ambiguity: C08 is a separate post-result follow-up to V2, not the V2 result itself. The C08-to-M10 relationship and full package checksum audit remain open; C08 must remain supplementary and must not support an M10-linked claim until those source records are available and reconciled.

## Supplementary Methods S1 — Retrospective portfolio selection

This integration is a retrospective, source-linked synthesis, not a systematic review or prospectively unified biological experiment. The main-text cases were selected to illustrate distinct evidence transitions and include both supported and unsupported outcomes; this selection was made with knowledge of earlier project outcomes and is not an outcome-independent sample of the portfolios. The initial synthesis and M2/M5 tests were followed by a post-result DMP fast-slow  $\times$  correspondence module and a source-informed V1 PC $\rightarrow$ SST figure/ground package. Each local protocol was frozen before its own outcome run, but all were designed after earlier project results had been inspected; none is project-level preregistration or independent biological confirmation. The V1 protocol's AP interaction interpretation had a sign-direction wording error, preserved and disclosed in its post-run amendment. Each Route-D portfolio record retains its original evidence class, unit, task, comparator, status (including invalid, blocked, or outcome-informed), and correction history in Table S1. Selected branch records are accounted for separately in Table S2; unresolved C08-to-M10 identity is not used to support a claim. A record's presence in the portfolio matrix does not imply that it supports the central claim. Supplementary-only and excluded records remain visible in the disposition accounting, including RR18/RR19 as separately governed exclusions. No common effect or meta-analysis is calculated because the tasks, outcomes, units and sampling structures are not exchangeable.

## Supplementary Methods S2 — Source provenance and reproducibility boundary

The NeuroMotif public main candidate is 1adc07ffc735cf458116ca36c9f4875708034d4a. The NeuroMech public experiment-publication candidate is 3e5c458f15b2cad7aa6c5703b1827763e84343bb; its local worktree had two changed/untracked paths at audit. The broader local Route-D view is based on 555c064f4ff3153c871808c3f191b9934a18eaf5 plus 387 changed/untracked paths. These states are not interchangeable and none is an owner-approved immutable full freeze. Paper-critical file hashes are in `./workstreams/PAPER_CRITICAL_SOURCE_SNAPSHOT.json`; this proves file identity at audit time, not that a result was independently rerun or redistributable.

Source files and historical environments remain heterogeneous. Exact runners, dependency locks, licenses and redistribution rights must be evaluated per selected record. The integration repository is not a replacement for either source repository and does not claim full independent reproducibility.

## Supplementary Methods S3 — Cerebellar source-readout reanalysis

This post-result exploratory analysis used the released Dryad v4 file `learning_1s_to_2s_GrC_CF.mat` [10] and source-code-derived plasticity computation. All sessions in the three released groups were included: five 1-s expert sessions, five 2-s novice/1-s expert sessions and six 2-s expert sessions. Eligible rewarded trials satisfied the source `mvlen > 7` and `goodmvd` criteria. Each session used five deterministic random 50/50 trial splits, with seed `20261001 + 100000g + 1000s + k` using zero-based group, session and split indices. Climbing-fibre selection, activity scaling and readout weights were fitted only on training trials.

Candidate climbing fibres showed a reward-aligned event increase in  $[0, 0.25]$  s relative to  $[-0.3, -0.025]$  s. The source plasticity computation used granule-cell activity in  $[-0.15, -0.025]$  s before teaching events, with rounded frame offsets and inclusive window endpoints. Activity was rectified, scaled by training-cell 95th percentiles, transformed with the source logistic function, averaged and normalized per selected climbing fibre, then averaged with the source negative sign. These weights were modeled proxies under the inherited connectivity assumption. Held-out readouts used movement-midpoint-aligned activity through the recorded reward delay, capped at 2 s, and squared Pearson correlation with elapsed time. Uniform, cell-shuffled and timing-shuffled controls and a training-standardized ridge reference with  $\alpha = 1.0$  were evaluated on the same held-out trials. Valid trial scores were averaged within split, splits within session and sessions within source group. Two splits without training climbing-fibre candidates remained unusable. No animal-level inference was calculated, and the ridge reference was not capacity matched.

The exact package is pinned to NeuroMech experiment-publication commit 3e5c458f15b2cad7aa6c5703b1827763e84343bb and selected-file hashes in `./reproducibility/SELECTED_ARTIFACTS.csv`. Route-D V3 has no exact record for this package and classifies the broader M5\_CEREBELLAR line as supplementary-only; this analysis is therefore reported as a separate supplementary branch record and is not included in the 39-row Route-D disposition counts.

## Supplementary Methods S4 — DMP fast–slow correspondence V1

The DMP V1 synthetic extension was motivated by the published fast–slow dual-memory spiking architecture of Sun et al. [13], but was not a reproduction of its algorithm or hardware. Its artificial model combined 32 leaky integrate-and-fire units (leak 0.5, threshold 1, subtractive reset and fixed fast-sigmoid surrogate gradient) with a fixed two-dimensional LMU-like state. The Legendre-derived continuous matrices were discretized by exact zero-order hold at horizon  $\theta = 32$ . The short-state control retained the same state dimension, fixed input matrix, learnable tensor shapes, initialization scheme, optimizer and update count, but used a 0.5I transition. Slow-state output entered the membrane current through a learned scalar-compressed input. DMP and control parameter counts were 1,286/1,385 for regression/classification; an 18-unit GRU reference had 1,315/1,372 parameters, within 5% of DMP. FLOPs were not matched.

Each generator used four input channels and a 64-step training sequence. In delayed association, a noisy four-level context occupied steps 0–3 and a noisy four-level query occupied steps 36–39; a fixed task-specific permutation of (context + query) mod 4 supplied the class. For state estimation, a stationary unit-variance AR(1) latent trajectory ( $\phi$  sampled from 0.90–0.98 per task instance) was observed through two channels; a binary context indicated which channel had low (SD 0.25) or high (SD 1.2) noise, and target was the final latent state. Context regime switch probability was sampled from 0.015–0.045. In contextual decision, a four-level context at steps 0–3 selected a task-specific mapping from four noisy evidence votes at steps 36–51 to a class. Longer-delay test sequences had 128 steps: query or evidence moved to steps 68–71 or 68–83, respectively; state-estimation horizon doubled. Configuration seeds were shared across IID and longer-delay generator instances, while trial draws differed.

Each task-instance seed generated 2,048 training, 512 IID test and 512 longer-delay test trials. Training prefixes  $N = 128, 512$  and 2,048 were nested. For every prefix separately, the slow-context stream was reassigned by a no-fixed-point derangement within that prefix; IID and longer-delay splits used independent derangements. This preserved the exact context-stream marginal for each comparison while holding the fast stream, targets and sample count fixed. The intervention broke trial-level pairing and did not change target labels. Two paired initializations were averaged within each task-instance seed.

All learned models used ten fixed epochs, batch size 64, Adam learning rate 0.003 and gradient clipping at five. Within each  $N$ , factorial cells used the same task data, update count and paired initializations. Across  $N$  in the secondary scaling analysis, fixed epochs meant that update counts increased with sample size: 20, 80 and 320 updates per fit at  $N=128, 512$  and 2,048. The three families used 30 task-instance seeds each. Classification endpoint was test 0–1 loss; state endpoint was test squared error against the unit-variance latent state. The co-primary contrast was  $I = [L(M+, C-) - L(M+, C+)] - [L(M-, C-) - L(M-, C+)]$ , where  $M+$  is DMP,  $M-$  is the short-state control,  $C+$  is aligned and  $C-$  is broken correspondence. Each family had a paired task-seed percentile bootstrap 95% interval and two-sided sign-flip  $P$  value; Holm correction controlled the three co-primary interactions. Task viability required aligned, full-data GRU loss to be below task-null loss; the three viability  $P$  values were also Holm-adjusted. A positive mechanism-specificity gate required a passed viability gate, positive adjusted interaction evidence and a positive aligned DMP advantage. The frozen secondary scaling criterion estimated the within-unit slope of aligned control-minus-DMP loss against  $\log_2(N)$ ; its three family  $P$  values were Holm-adjusted. It required the  $N=128$  advantage interval above zero, slope interval below zero and adjusted  $P < 0.05$ . Because optimizer update count varied with  $N$ , this criterion did not isolate sample efficiency. Longer-delay changes were secondary and evaluated without refitting.

The runner, verifier and contract hashes and runtime environment were frozen before confirmatory units. Runtime was Python 3.9.6, NumPy 2.0.2 and PyTorch 2.8.0 on macOS arm64 CPU, deterministic algorithms enabled and one thread. The run produced 7,020 unit-level rows and 7,020 per-trial loss vectors. The independent verifier recalculated all means from raw vectors and checked hashes, task-instance coverage, parameter counts, cell completeness and exact correspondence marginals. The module is a post-result project extension because prior M2/M5 outcomes and the fast–slow literature had been reviewed before it was designed.

## Supplementary Methods S5 — Prospective source-informed V1 figure/ground test

The source paper reports like-to-like functional connectivity from orientation-tuned pyramidal cells to SST interneurons and a causal contribution of SST activity to orientation-dependent surround suppression [14]. The present benchmark did not use recordings from that study. Each  $11 \times 11$  synthetic image contained eight orientation-tuned response channels. A fixed soft tuning function ( $\kappa=2$ ) generated channel responses with Gaussian noise (s.d. 0.08), positive clipping and normalization. One randomly placed  $3 \times 3$  figure had an orientation category selected uniformly from the seven alternatives to the background category. Training and IID backgrounds were uniform; the OOD split used the fixed category probabilities [0.35, 0.25, 0.12, 0.08, 0.07, 0.05, 0.04, 0.04]. Each of 30 deterministically derived task-instance seeds generated 512 training images, 256 IID test images and 256 OOD images. Nested training prefixes were 32, 128 and 512 images.

At each location, surround activity was the mean of the eight neighbouring locations. The feature-selective transform subtracted 0.65 times the same-channel surround response. In the broken-correspondence arm, channels were routed through a fixed cyclic no-fixed-point permutation. The historical global-pool control subtracted 0.65 times the across-channel mean surround response. Normalization makes each response vector sum to one; its neighbour mean also sums to one, so the pooled scalar is  $1/8$  in exact arithmetic. This control is a constant offset,  $x - 0.65/8$ , absorbed by the class-mean-centered linear

readout and its intercept. It is effectively local-only, up to floating-point rounding, and not a second functional contextual inhibition mechanism. Global aligned/broken cells are identical by construction; the interaction reduces to selective broken minus selective aligned. Frozen cell names are retained in source files. A regularized linear discriminant (ridge  $\lambda=0.02$ ) was fit separately in each factorial cell using the same training images. A generic-context reference received concatenated center and surround channels (16 features) with the same readout; it was not parameter-matched and was not a strong neural-network reference. The frozen prevalence reference was 9/121. AP used stable descending pixel sorting and averaged precision at each positive pixel rank; equal-score ties retained flattened pixel-index order. Thus the actual all-equal-score AP can depend on patch position and is not identically 9/121. Passing the frozen prevalence-reference criterion does not establish contextual learning or rule out structure-preserving no-learning alternatives.

The independent unit was the task-instance seed; image and pixel measurements were nested. The primary endpoint was mean per-image average precision. The frozen interaction was  $(AP(\text{selective, broken}) - AP(\text{selective, aligned})) - (AP(\text{global, broken}) - AP(\text{global, aligned}))$ . The contract text incorrectly described a positive value as an alignment benefit even though AP is higher-is-better and the contrast is broken-minus-aligned; the numerical formula, outputs and gate were left unchanged. The positive-specificity gate required generic-reference viability, the frozen numerical interaction criterion and higher aligned AP for selective than global pooling. Paired bootstrap CIs used 20,000 task-seed resamples; two-sided sign-flip tests used 100,000 Monte Carlo sign patterns generated by NumPy default\_rng(20261003). Scaling and OOD comparisons were secondary. The executed contract, runner, verifier, derived seed schedule, hashes, 230,400 saved per-image AP outcomes and interpretation amendments are in `../new_experiments/PROSPECTIVE_V1/`.

## Supplementary Results S1 — Full portfolio interpretation

Consult Table S1 for the complete row-level status and ceilings. In brief, the route contains positive, negative, invalid, inconclusive, schema-blocked and supplementary-only records. M2 comprises multiple synthetic/controller rounds and cannot be summarized by the single favorable sensory-site versus output-persistence contrast alone. M5 and related eligibility studies are heterogeneous; V3 classifies the M5 line as supplementary-only, while some selected experiment-publication packages have no exact V3 row. M11 and M12 must be retained in the full portfolio account, as must the *Drosophila* synthetic counterevidence and robustness records. These records constrain project-level selection but do not become evidence for a biological mechanism merely by being included.

## Supplementary Table S3 — M2/M5 outcome-informed synthetic V1 tests

M2 and M5 are separate synthetic studies with different outcomes and inferential units. M2 used 32 paired training-seed blocks and mean squared tracking error (MSE); its primary contrast is action-only minus realized-state MSE under hidden reversals, so a negative value favours action-only. M5 used 30 task seeds and normalized mean absolute error (nMAE); its contrast is broken minus aligned correspondence, so a positive value means error increased when input–target correspondence was broken. Intervals resample the corresponding paired seed units. No pooled estimate is defined.

| Study and arm                                               | Independent unit (n)     | MSE action-only / realized-state or<br>nMAE aligned / broken | Contrast and 95% interval                                             | Interpretation                                                                                                                                                                                                                       |
|-------------------------------------------------------------|--------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M2 primary: action-only vs realized-state, hidden reversals | Training-seed block (32) | MSE 2.43809 / 2.50618                                        | Action-only – realized-state = $-0.068094$ [ $-0.141077, -0.003541$ ] | The exact execution-state channel did not improve this controller; the contrast favoured action-only for this plant and training budget. It does not identify an information-specific penalty or test the biological RIM→AIY signal. |
| M5 CF-timed local rule                                      | Task seed (30)           | nMAE 0.141521 / 0.258109                                     | Broken – aligned = $+0.116588$ [ $+0.115072, +0.118065$ ]             | Error increased when correspondence was broken; the contrast was positive in all 30 seeds. This supports use of correspondence by the tested synthetic rule, not a biological mechanism.                                             |
| M5 full-input replay                                        | Task seed (30)           | nMAE 0.071385 / 0.260936                                     | Broken – aligned = $+0.189552$ [ $+0.186912, +0.192229$ ]             | The correspondence effect was larger than the local-rule effect; replay receives all input bins.                                                                                                                                     |
| M5 generic ridge on the same 16 features                    | Task seed (30)           | nMAE 0.056643 / 0.259033                                     | Broken – aligned = $+0.202390$ [ $+0.200123, +0.204675$ ]             | The generic feature-based effect was larger than the local-rule effect. The scalar ridge head, objective and fitting procedure differ from the local rule, so this is not a capacity-matched learning-rule comparison.               |
| M5 no-trace control                                         | Task seed (30)           | nMAE 0.258153 / 0.258083                                     | Broken – aligned = $-0.000070$ [ $-0.000411, +0.000219$ ]             | The point estimate slightly decreased after correspondence was broken; its interval crossed zero and no equivalence margin was specified.                                                                                            |
| M5 CF-time-shuffled control                                 | Task seed (30)           | nMAE 0.259511 / 0.258312                                     | Broken – aligned = $-0.001199$ [ $-0.002510, +0.000072$ ]             | The point estimate slightly decreased after teaching-time shuffling; its interval crossed zero and no equivalence margin was specified.                                                                                              |

The full seed-level outputs, frozen contracts, source/code provenance and independent verification records are in `../new_experiments/`. The M2 and M5 results are not pooled and do not validate either motivating biological mechanism.

## Supplementary Table S4 — DMP fast–slow × correspondence V1

The new module was a post-result project extension with a protocol frozen before its 90 task instances were generated. Its fast–slow abstraction was inspired by, but is not a reproduction of, a previously published dual-memory spiking architecture [13]. Each family had 30 independent task-instance seeds; the  $2 \times 2$  mechanism-by-correspondence interaction was the co-primary outcome. Positive aligned advantage means lower DMP loss than the matched short-state control. Task gates test whether the full-data aligned generic GRU beats the task-null predictor. Values below are recalculated from per-trial arrays by the independent verifier.

| Task family         | Interaction (95% paired task-seed CI) | Holm-adjusted P | Aligned DMP advantage at N=2,048 (95% CI) | GRU task gate | DMP aligned longer-delay loss change (95% CI) | Mechanism-specificity gate          |
|---------------------|---------------------------------------|-----------------|-------------------------------------------|---------------|-----------------------------------------------|-------------------------------------|
| Delayed association | -0.00212 [-0.00615, 0.00173]          | 0.64755         | +0.00208 [-0.00345, 0.00814]              | Fail          | +0.00143 [-0.00866, 0.01081]                  | Not met; viability gate failed      |
| State estimation    | -0.00218 [-0.00383, -0.00054]         | 0.05058         | -0.00243 [-0.00363, -0.00123]             | Pass          | +0.00096 [-0.00187, 0.00375]                  | Not met; direction favoured control |
| Contextual decision | +0.00101 [-0.00537, 0.00778]          | 0.77924         | +0.00156 [-0.00573, 0.00931]              | Pass          | +0.22096 [+0.20153, +0.23841]                 | Not met; interaction not detected   |

No family met the positive mechanism-specificity criterion. The state-estimation adjusted P value was 0.05058 and its estimate was negative; it did not meet a 0.05 threshold in the prespecified positive direction. The contextual-decision secondary scaling criterion, labelled a sample-efficiency gate in the frozen protocol, did pass: aligned advantage was +0.07236 at N=128 (95% CI, +0.05036 to +0.09557), +0.03506 at N=512 (+0.01335 to +0.05954), and +0.00156 at N=2,048 (−0.00563 to +0.00941); slope was −0.01770 per  $\log_2(N)$  step (95% CI, −0.02288 to −0.01274; Holm-adjusted P = 0.000060). This secondary, single-family result did not satisfy the primary mechanism-specificity gate and does not establish sample efficiency: with ten epochs and batch size 64, the three N values give 20, 80 and 320 optimizer updates per fit. Longer-delay differences are descriptive and are not interpreted as broad OOD generalization.

The complete 7,020-row metrics file, 7,020 raw loss vectors, task-instance configurations, pre-run freeze and verification hashes are retained in `../new_experiments/DMP_FAST_SLOW_CORRESPONDENCE_2X2_V1/results/`. The independent verification status is PASS.

## **Supplementary Table S5 — Prospective V1 PC→SST synthetic figure/ground test**

The study was selected after earlier project outcomes and frozen before its own single confirmatory run. AP is higher-is-better; the primary interaction formula below is broken minus aligned, so positive values indicate a relative advantage for broken routing under selective pooling. The contrast's interpretation in the frozen contract narrative had the sign reversed. Intervals resample the 30 task-instance seeds. The full raw per-image AP output and verifier-derived task-seed means are retained in `../new_experiments/PROSPECTIVE_V1/results/`.

| Endpoint/contrast                                           | Estimate    | Paired 95% CI        | Two-sided sign-flip P | Interpretation                                                                    |
|-------------------------------------------------------------|-------------|----------------------|-----------------------|-----------------------------------------------------------------------------------|
| Generic context minus prevalence null, N=512 IID            | +0.37219 AP | +0.36046 to +0.38379 | 0.000010              | Frozen prevalence-reference criterion passed; contextual learning not established |
| Selective minus constant-offset control, aligned, N=512 IID | −0.02208 AP | −0.02737 to −0.01664 | 0.000010              | Aligned selective routing was worse than the effectively local-only control       |
| Selectivity × correspondence, N=512 IID                     | +0.01603 AP | +0.00815 to +0.02357 | 0.000600              | Opposite to the stated alignment-benefit prediction                               |
| Selectivity × correspondence, skewed-background OOD         | +0.01692 AP | +0.00553 to +0.02913 | 0.007250              | Secondary; same direction as IID                                                  |
| Selective-minus-global aligned slope vs $\log_2(N)$         | +0.00091 AP | −0.00108 to +0.00291 | 0.376247              | No reliable sample-size trend                                                     |

The numerical interaction criterion passed. Its positive broken-minus-aligned AP direction contradicted the prose prediction of an alignment benefit. Separately, the conjunctive gate failed because aligned selective routing underperformed the constant-offset, effectively local-only control. These facts do not change the frozen procedural decisions. The stored verifier returned `integrity_pass=true`: 230,400 saved per-image AP rows, 180 seed  $\times$  sample-size  $\times$  split groups, frozen hashes and global-cell permutation invariance were checked. It recomputed means and inference from saved AP, not response features or AP from prediction scores. This is integrity plus endpoint/statistical recomputation, not full end-to-end reconstruction. The original contract's stronger verification commitment remains a disclosed deviation. The post-run amendment records the sign-direction wording error and the limited linear comparator. The experiment used generated population activity rather than neural observations, and the source paper's Zenodo archive was not downloaded or analysed.

## Supplementary Table S6 — Comparator matching and claim limits

Matching is specific to the contrast being interpreted. “Not measured” means that the retained protocol/output does not establish a compute match; similar wall-clock time or parameter count must not be substituted for a FLOPs or inference-compute audit.

| Study / comparison                              | Input information                                                                                                                                                                                                    | Trainable capacity                                                                                                                                | Training / optimization budget                                                                                                                                                                                                        | FLOPs or inference compute                                                                                 | Extra task knowledge                                                                                                          | Defensible comparison                                                                                                                                                                        |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M2 V1, action-only vs realized-state AcRKN      | Same six-channel interface; the auxiliary channel is invalid/zero for action-only and carries the previous executed-displacement sign for realized-state. All remaining observations, flags and commands are shared. | Matched at 320 trainable parameters per AcRKN arm.                                                                                                | Matched: 160 Adam updates, same learning rate, batch size, sequence length, seed pairing and environment streams.                                                                                                                     | Not profiled or matched.                                                                                   | Neither learned arm receives future/current unexecuted reversal information.                                                  | Isolates the added realized-state channel within this AcRKN implementation and budget. It does not establish an intrinsic information penalty or test the biological RIM→AIY signal.         |
| M2 V1, AcRKN vs GRU reference                   | Both learned families receive the same six inputs.                                                                                                                                                                   | Near-count matched, 320 vs 323 parameters; architecture differs.                                                                                  | Same update count and training stream protocol.                                                                                                                                                                                       | Not profiled or matched.                                                                                   | No explicit task-distribution oracle in either learned model.                                                                 | Contextualizes performance; does not isolate an architecture effect.                                                                                                                         |
| M2 V1, learned models vs particle filter        | The particle filter uses the task's observation/actuation distributions; in the revealed condition it also receives preceding reversal events.                                                                       | Not capacity matched; particle count is not comparable to trainable parameter count.                                                              | Particle-filter gains selected on disjoint validation seeds; learned models use gradient training.                                                                                                                                    | Inference compute not matched; particle counts 64–256 were examined, with 256 used in the main evaluation. | Yes. It uses known task dynamics/distributions.                                                                               | A task-aware reference point/context only, not a fair architecture or parameter comparison.                                                                                                  |
| M5 V1, CF-timed local rule vs generic RBF ridge | Same 16 temporal RBF features plus bias.                                                                                                                                                                             | Not matched: local softmax rule has 476 parameters; scalar ridge has 17. The frozen “parameter-matched” label was corrected to “feature-matched.” | Not matched: local rule updates once per trial after complete-trial feature construction; ridge is a closed-form fit with a different scalar regression objective. The local rule is not an online within-trial trace implementation. | Not profiled or matched.                                                                                   | Both use the same supervised targets; neither is given biological observations.                                               | Shows that a generic learner can exploit the same feature representation more effectively in this task. It cannot identify a learning-rule-specific advantage.                               |
| M5 V1, CF-timed local rule vs full-input replay | Local rule uses compressed 16-basis full-trial features; replay receives all 40 input bins.                                                                                                                          | Not matched: 476 vs 41 fitted parameters.                                                                                                         | Not matched: one-pass local updates vs closed-form fitting.                                                                                                                                                                           | Not profiled or matched.                                                                                   | Replay has access to the complete input trial and target labels during fitting.                                               | A stronger-information algorithmic reference; the larger correspondence effect does not isolate mechanism identity.                                                                          |
| DMP V1, DMP vs short-state control              | Same task inputs, targets and context stream; aligned/broken correspondence is crossed within each architecture.                                                                                                     | Matched within the factorial contrast: same learnable parameters, input matrix and state dimension; transition dynamics differ.                   | Matched within study: paired initializations, ten epochs, batch size 64, same Adam rate and task-seed protocol.                                                                                                                       | FLOPs explicitly not matched.                                                                              | No external task oracle is specified for this model contrast.                                                                 | The primary interaction tests whether the tested slow-state transition changes sensitivity to correspondence, within these task generators. It does not prove biological mechanism identity. |
| DMP V1, DMP vs generic GRU viability reference  | Same task families and target definitions; architecture-specific internal states differ.                                                                                                                             | Near-count matched within 5%.                                                                                                                     | Same epochs, batch size, learning rate and paired-initialization scheme.                                                                                                                                                              | FLOPs not matched.                                                                                         | GRU is a generic task-learning reference; the task gate compares its aligned full-data performance with a task-null baseline. | Viability/context check, not a compute-matched superiority benchmark.                                                                                                                        |

| Study / comparison                                           | Input information                                                                                                                                                                                                                                                              | Trainable capacity                                                                    | Training / optimization budget                                                                                | FLOPs or inference compute | Extra task knowledge                                                                         | Defensible comparison                                                                                                                                              |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prospective V1, selective vs global pooling factorial cells  | Same generated images; selective routing uses neighbour channel information. Normalization makes global pooling a constant 1/8, effectively removing variable surround information from that control. It is a constant-offset/local-only equivalent, invariant to permutation. | Matched among four factorial cells by the same eight-input pathway and ridge readout. | Same ridge-discriminant fit and $\lambda = 0.02$ across factorial cells.                                      | Not profiled or matched.   | No source neural recordings or external task oracle; all arms use synthetic training labels. | Tests the specified selective transform against a degenerate, effectively local-only control. It does not compare two functional contextual inhibition mechanisms. |
| Prospective V1, factorial cells vs generic-context reference | Generic reference concatenates local and surround channels (16 inputs); factorial pathway uses eight pooled/routed channels.                                                                                                                                                   | Not matched.                                                                          | Same ridge-regularized linear-discriminant fitting family and $\lambda$ , but feature dimensionality differs. | Not profiled or matched.   | Same synthetic task labels; the reference is linear, not a strong neural-network comparator. | Passes the frozen prevalence-reference criterion only. The gate does not establish contextual learning; the reference is not a matched superiority benchmark.      |

## Supplementary Table S7 — Inferential hierarchy across the selected evidence classes

| Study                                         | Primary estimand / gate                                                                                                  | Independent unit                                                                 | Multiplicity and inferential status                                                                                                                                                                                                                 | Claim ceiling                                                                                           |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| M0 workstream: recovered Step-2/3/3A sequence | Nonzero interference with source $R^2 \geq 0.30$ , recurrent > passive retention and $\geq 6/8$ seeds in that direction. | Eight paired weight-initialization seeds per species $\times$ interference cell. | Recovered calibration/falsification outputs/configuration; conjunction not met. Planned M0 pilot and T00–T11 main experiment are not documented as completed. Stored-output means and direction counts recomputed; no raw-to-result reconstruction. | Bounded historical sequence only; not a completed M0 pilot or field-wide topology conclusion.           |
| Fish1.5                                       | Positive recurrence–persistence association; separate frozen topology null.                                              | One specimen with 82 neurons; neurons are conditional subsamples.                | Primary directional association; neuron resampling does not capture all dependence. Frozen topology null invalid (861/1,000 undefined). Secondary procedures remain separate.                                                                       | Conditional descriptive association and invalid-null record, not animal-population or causal inference. |
| Worm source reanalysis                        | Planned animal/session-aligned computation test.                                                                         | Required animal/session/stimulus keys unavailable in the released schema.        | Blocked; no eligible confirmatory endpoint.                                                                                                                                                                                                         | No evidence against the published source computation.                                                   |
| Cerebellar source readout                     | Modeled CF-timed weights versus readout alternatives within sessions.                                                    | 16 source sessions; within-session held-out splits.                              | Post-result descriptive source-data analysis; not a biological replication or animal-level causal test.                                                                                                                                             | Session-level prediction with modeled weights; no synaptic-mechanism or transfer claim.                 |

| Study          | Primary estimand / gate                                                                                                                     | Independent unit                                                  | Multiplicity and inferential status                                                                                                                                                                                                                                                                                                                                                                                                           | Claim ceiling                                                                                                                    |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| M2 V1          | Action-only minus realized-state MSE under hidden reversals; positive would favour the added realized-state input.                          | Paired training-seed block (n = 32); episodes nested.             | The primary interval excluded zero in the direction favouring action-only. Paired bootstrap; primary and diagnostic intervals unadjusted, with no portfolio-wide error control or equivalence margin.                                                                                                                                                                                                                                         | A bounded result for this AcRKN, plant and training budget; diagnostics do not isolate an intrinsic information penalty.         |
| M5 V1          | CF-timed local-rule nMAE(broken) minus nMAE(aligned); positive means breaking correspondence increases error.                               | Task seed (n = 30); trials nested.                                | Task viability passed and primary correspondence effect was positive in all 30 seeds. Paired-seed bootstrap; comparator contrasts are descriptive secondary results and are not multiplicity adjusted.                                                                                                                                                                                                                                        | Correspondence mattered to this synthetic rule/task; comparator results do not establish learning-rule specificity.              |
| DMP V1         | Three task-family-specific mechanism $\times$ correspondence interactions; separate task-viability gate; secondary small-data scaling gate. | Task-instance seed within family (n = 30); trials/repeats nested. | No family passed the positive mechanism-specificity gate. Holm correction across the three co-primary interactions, separately across three viability tests; scaling slope has its prespecified Holm-adjusted gate. Longer-delay results are secondary/descriptive.                                                                                                                                                                           | No family passed the mechanism-specificity criterion; the contextual-decision small-data pattern is a narrower secondary result. |
| Prospective V1 | Selectivity $\times$ correspondence AP interaction plus a conjunctive positive specificity gate.                                            | Task-instance seed (n = 30); images/pixels nested.                | The frozen prevalence-reference criterion and numerical interaction condition passed. The interaction opposed the prose alignment-benefit prediction; aligned selective routing underperformed the constant-offset/local-only equivalent control, failing the conjunctive gate. Contextual learning was not established. Secondary sample-size/OOD endpoints are not pooled with the primary or corrected across the wider project portfolio. | Synthetic task only. The positive specificity gate failed; do not recode the interaction as an aligned-transfer benefit.         |

These evidence classes do not share an estimand, unit, endpoint scale or multiplicity family. Invalid and blocked rows do not contribute valid negative mechanism evidence. The intervals and P values must therefore be read within each frozen study contract, not ranked across studies or interpreted as a pooled evidence score.

## Supplementary Results S2 — Historical artificial tests and source-readout record

The following numerical account was moved from the earlier main Results or retained from historical source packages. These studies retain their original identities and ceilings; they are not replications of the completed V1 tests. Supplementary Table S2 distinguishes exact hash-pinned branch packages from unresolved cross-branch record relationships. The source-readout endpoint was held-out trial squared Pearson correlation with time progress, a direction-insensitive association rather than calibrated time prediction. Of 80 planned splits, 78 were usable; two retained the status NO\_TRAINING\_CF\_CANDIDATES. Source weights were modeled and sessions were not established as independent animals.

In one synthetic switching task, sensory-site motor feedback reached 88.94% accuracy versus 82.14% for an equal-parameter output-persistence control, but did not exceed no-feedback (89.05%), a two-unit generic recurrent model (90.21%), or a task-aware Bayes filter (90.30%). The local contrast therefore identifies a placement-dependent trade-off against one comparator, not an overall accuracy advantage.

In the cerebellar source analysis, model-derived weights exceeded uniform weights in 15/16 source sessions and CF-time-shuffled weights in 16/16; a fixed ridge readout exceeded the source-derived projection in 14/16. This supports within-session predictive temporal structure under the modeled analysis. In the separate synthetic interval task, CF-timed LTD did not meet its frozen transfer criterion: mean held-out absolute error was 0.1252 s versus 0.1097 s for no-trace, 0.1252 s for CF-time-shuffled teaching, 0.1030 s for a generic radial-basis representation, and 0.0753 s for the training-label empirical timer. A design audit found that trial-specific timing jitter was independently sampled and not present in the inputs. The outcome bounds this implementation/task pair; it does not test whether the biological rule could use a suitable trial-specific signal.

Two additional M5 eligibility-trace studies use different synthetic objectives and must not be combined with the source-defined CF-LTD interval experiment. In a delayed-XOR classification task (32 seeds), eligibility traces reached 0.6974 accuracy versus 0.5562 for no-trace and one-step truncated backpropagation through time (TBPTT-1), but trailed TBPTT-4 (0.9633) and full backpropagation (0.9996). In a separate fixed-generator classification suite, trace eligibility exceeded no-trace by 0.2284 (95% interval [0.2213, 0.2357]); exact replay nevertheless outperformed eligibility in all 12 generator-by-delay cells (overall means 0.7609 versus 0.5590). These are post-result synthetic demonstrations that eligibility can help against a weak memory control under some task conditions; the stronger replay and gradient-based controls set a lower ceiling on claims. Neither result validates a cerebellar mechanism or establishes general transfer.

The M2, M5 source-readout, source-defined transfer, delayed-XOR and task-generator records are pinned to the NeuroMech experiment-publication commit 3e5c458f15b2cad7aa6c5703b1827763e84343bb. Existing stored-output verifiers passed for M2 and the two source-defined CF-LTD packages; stored verification records are available for the delayed-XOR and generator studies, and selected artifact hashes are recorded in `'SELECTED_ARTIFACTS.csv'`. These packages are post-result exploratory at the project level, and the source-readout summaries are descriptive. This pins the cited study records; it does not resolve the broader NeuroMech mainline/worktree freeze or which additional studies belong in the integrated paper.

## Supplementary Results S3 — Fish1.5 inference and M0 source boundary

Fish1.5 retained 10,000 valid positive-association permutation draws and 9,844 valid neuron-bootstrap draws out of 10,000 requested draws. The reported bootstrap interval summarizes the valid draws. The frozen topology null preserved binary in/out degrees and the global weight multiset, but not each node's weighted strength. With 861 undefined correlations, its P value remains unestimable. Post-result conditional null analyses do not repair the original frozen inference and retain their separate labels in Table S1. The M0 corrected report and canonical Step-3/3A artifacts have now been recovered into `../evidence/m0_source_archive/`. They document a calibration/falsification and stopping sequence under the M0 workstream label; the planned M0 pilot and T00–T11 main experiment are not documented as completed. Resolved configurations, seed-level outputs and verdicts identify eight paired weight-initialization seeds per species  $\times$  interference cell. No nonzero level met the conjunction of source  $R^2 \geq 0.30$ , recurrent M3 > passive M1 and at least six of eight seeds in that direction. Crossovers remained below the recoverability floor. Means and direction counts were recomputed from the saved CSVs. The manifests label the rule prospectively frozen, not preregistered. Exact processed input tensors were not archived, manifests had empty dataset lists, and the exact command and dirty execution diff remain unresolved. Thus claim-linked stored-output provenance is substantially restored, while end-to-end rerun provenance is open. See `../evidence/M0_PROVENANCE_AUDIT.md`; the M0 case does not become part of the NeuroMotif public snapshot by reassignment.

## Supplementary Results S4 — M2 V1 diagnostics and reproducibility

The full arm-by-condition means, primary and diagnostic intervals, parameter counts and repeated-run comparison are retained in `./new_experiments/M2_REALIZED_STATE_VS_ACTION_BELIEF_V1/RUN_REPORT.md` and `runs/canonical_1/summary.json`. All 32 seed blocks are included. Both full runs passed the independent verifier and agree scientifically after only declared timing fields are excluded. The revealed-event control had no frozen equivalence margin; its zero-crossing interval cannot establish equivalent performance. The benchmark received an exact execution-sign measurement, with no tested corruption or delay condition.

The M2 base HEAD `3e5c458f15b2cad7aa6c5703b1827763e84343bb` did not define the executed version: the captured contract was modified and the runner/verifier were untracked. `SOURCE_PROVENANCE.json` identifies the exact contract SHA-256 `489ed473bbeb895ccb99ad8b661dbbdeace0f2a19fc6615e1a4dfa8fd6736f0`, runner `984ef632108a0aa3256a19009747194818db1296ed0531fe1b294703cf27c8d6`, and verifier `f6d48727f4ece27f385d56baf1221ac12e09b0cb36ef6e52e6e47d867e7c7055`.

## Supplementary Results S5 — M5 V1 controls and comparator correction

Full means and effect intervals are retained in `./new_experiments/M5_CORRESPONDENCE_PRESERVED_TIMING_V1/RESULTS.md` and `results/primary_result.json`. The no-trace broken-minus-aligned nMAE effect was  $-0.000069515$  (95% interval  $[-0.000411054, +0.000219329]$ ); the teaching-time shuffle effect was  $-0.001199416$  ( $[-0.002510140, +0.000071742]$ ). Zero-crossing intervals do not establish equivalence or zero information. The same-feature ridge arm has a scalar 17-parameter head, whereas the local rule has a 476-parameter, 28-class softmax head; objectives and fitting procedures differ. The preserved frozen contract and post-run audit record the correction from parameter matched to feature matched. Teaching-time shuffling is a training intervention; the aligned/broken regime independently changes input-target pairing in both train and test splits, so the primary effect does not separate training and evaluation contributions.

The M5 contract SHA-256 is `1434055cc21047edfd8185d50e6dc207c640ddb46aa3cf7b1293a8353c0e7658`, and the successful runner SHA-256 is `9f9f4d810f473573fc86f8a8bb2ad454a4d099009004429618d7b766f3bbad80`. The first launch failed before producing outcomes because Python 3.9 did not support `zip(strict=True)`; a syntax-only amendment was frozen before the successful run. The original freeze, amendment, execution incident and successful run manifest remain intact.

## Supplementary Results S6 — DMP V1 interpretation and verification

Across the three families, no positive mechanism-specificity gate passed. The delayed-association task was marked non-interpretable for the mechanism claim because the aligned generic GRU did not beat the task-null predictor after the prespecified viability analysis. In state estimation, the DMP  $\times$  correspondence estimate was negative and its paired interval excluded zero in the negative direction; the Holm-adjusted P value (0.05058) was just above the two-sided 0.05 cutoff. The aligned DMP loss was also higher than the aligned short-state-control loss. This supports a task-bounded disadvantage of this DMP implementation in the tested generator, not a claim that slow state is generally harmful.

In contextual decision, aligned DMP showed a large relative advantage at  $N=128$  that diminished at  $N=512$  and was not detected at  $N=2,048$ . Its predeclared slope gate passed, but its full-data mechanism  $\times$  correspondence interaction and aligned advantage did not. Moreover, under the longer-delay shift its aligned loss rose by 0.22096 (95% paired task-seed CI, 0.20153 to 0.23841), whereas the delayed-association and state-estimation aligned changes were near zero with intervals spanning zero. This indicates sensitivity to the tested task/shift combination. All models used ten epochs, so  $N=128$ , 512 and 2,048 entailed 20, 80 and 320 updates per fit with batch size 64. The scaling result is conditional on this fixed-epoch protocol and does not isolate data quantity from optimization steps or establish general OOD robustness.

`RUN_MANIFEST.json` records 30 seeds per family and 7,020 metrics/raw loss vectors. `VERIFICATION.json` records PASS; file hashes, exact marginals, metric recomputation and complete cells passed. The results remain synthetic and post-result at the project level. Fast-slow memory already has direct prior art, including a 2026 *Nature Machine Intelligence* dual-memory spiking model [13].

## Supplementary Results S7 — Prospective V1 interpretation and verification

The source-informed synthetic task passed the frozen prevalence-reference criterion, but that criterion does not establish contextual learning. Feature-selective aligned routing had lower AP than the constant-offset, effectively local-only control produced by the normalized global-pool transform. The frozen interaction was statistically different from zero in the direction indicating better performance after correspondence breaking; under the contract's formula and higher-is-better AP, this is opposite to the written alignment-benefit prediction. Therefore, the package does not support positive mechanism specificity or an aligned-routing utility advantage. The same interaction direction appeared under the fixed skewed-background test, while the sample-size slope was inconclusive.

An audit found that the contract's positive interaction interpretation had the sign reversed. Because the issue was identified after outcome generation, no contract, output, model or gate was changed. The numerical interaction condition passed; its scientific direction opposed the prose prediction. The conjunctive gate failed on the aligned selective-minus-control condition. The original direction amendment and the later TECHNICAL\_AUDIT\_AMENDMENT.md preserve the frozen formula, gates and outputs while correcting control algebra, reference interpretation and verification scope. The selected source/task package is new to the tested mechanism portfolio but was selected after earlier project outcomes and uses the visual modality already present in adjacent synthetic-visual work. It is not an independent team replication or biological confirmation. The generic context reference is linear and not a strong neural-network alternative.

A separate post-audit implementation independently rebuilt the first frozen V1 task seed at N=512 on the IID split. It regenerated inputs, transforms, fitted readouts, scores and AP without importing the experiment runner. In the frozen Python 3.12.13 / NumPy 2.4.4 runtime, all 1,280 AP values across five conditions exactly matched the archive. This is bounded Level-3 evidence for 5/900 endpoint cells, not full reconstruction. Global and independently fitted local-only scores differed by at most  $8.77 \times 10^{-9}$ . Across these 256 masks, all-equal-score stable-tie AP ranged from 0.05630 to 0.18696 and averaged 0.10231, rather than the prevalence reference 0.07438. These audit diagnostics do not replace any frozen endpoint or gate and do not establish a no-learning explanation for the observed learner performance. Scores, masks and code are retained in `../reproducibility/V1_ENDPOINT_AUDIT/`.

## Supplementary Methods S6 — IRCN budget-matched scheduling pilot

The IRCN line was reviewed at branch v0.7.0-runtime-icef-bridge, commit 8eac42d. Its v0.8.0 CBCC pilot directory is present in that working tree and is not committed at that head. IRCN's provenance record classifies NeuroConverge and nmi\_feedback\_routing as existing adjacent workspaces under a read-only boundary, not used as IRCN inputs, and its v0.7.0 development contract excludes connectome input, NeuroConverge modification and NMI manuscript edits. The inclusion of the line in this manuscript is therefore an owner-authorized scope amendment, recorded here so that neither declared boundary is rewritten; IRCN's earlier conclusions are not used as evidence inside IRCN itself.

Sources used: the v0.8.0 CBCC development and pilot report; the frozen second preregistration; the pilot run directory of 9 October 2026; the two validation-gate records; the v0.7.0 completion record and decision; the work-package status table; the project decision record; and the protocol-deviation record. Every quantity below was recomputed from pilot\_seed\_metrics.csv and matched against confirmatory\_freeze.json to a tolerance of 1e-12. Per-artifact SHA-256 hashes are recorded in evidence/IRCN\_EVIDENCE\_DISPOSITION.json.

### Frozen eligibility rule, transcribed from the pilot driver

For each control policy p and each pilot seed s:

```
```text
mismatch(p, s) = |MAC_p(s) - MAC_treatment(s)| / max(MAC_treatment(s), 1)
p is eligible <=> mismatch(p, s) <= 0.01 for EVERY pilot seed s
```
```

The denominator is the treatment's own realized operation count, which is produced by running the treatment. A control's admissibility is therefore a function of the treatment's realized behaviour rather than of the plan.

## Supplementary Table S11 — IRCN pilot eligibility outcome

| family          | treatment MAC per seed | control mismatch range | eligible controls | frozen confirmatory state           |
|-----------------|------------------------|------------------------|-------------------|-------------------------------------|
| sparse_key      | 500,288 – 508,160      | 3.79 % – 15.74 %       | none              | baseline unset; 0 seeds; infeasible |
| feedback_switch | 25,344                 | 1466.92 % – 1987.63 %  | none              | baseline unset; 0 seeds; infeasible |

Frozen record: sample\_size\_feasible = false; confirmatory test-seed list empty; no confirmatory evaluation run. Descriptive mean losses are reported in the main text as descriptions only; no Holm-adjusted comparison was computed.

### **Supplementary Results S8 — IRCN gates that passed without licensing a utility claim**

The line's correctness gate passed 96/96 calibration cases and 240/240 untouched-seed cases with a maximum NRMSE of  $4.2635\text{e-}9$  against a limit of  $6.8376\text{e-}8$ , and its decision record still declines an efficiency claim and declines to enter the baseline comparison. Its runtime bridge reproduced the frozen synchronous cell on 21 fixtures with 42 of 42 repeated attempts passing and a maximum state difference of  $2.8311\text{e-}15$ , and its decision record forbids an efficacy claim and the next evaluation stage. Its cost calibration found no threshold at or below the frozen error ceiling, leaving its results table header-only, and carries a confirmed deviation because the smooth-input event cadence was set by implementation convention rather than frozen before the run. Work-package status is WP0 complete with audit limitations, WP1 pass for accuracy and correctness scope only, WP2 inconclusive, WP3 to WP6 blocked.

### **Supplementary Table S8 — Related work and bounded contribution**

This is a targeted comparison of closely related work, not a systematic novelty survey. Source-specific summaries are limited to the claims identified below; the author-facing source record is in `../evidence/NEAREST_NEIGHBOUR_COMPARISON.md`.

| Prior work                                               | What it establishes                                                                      | Present question and increment                                                                                                        | Limit of the distinction                                                                                                     |
|----------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Suárez et al., conn2res [1]; Beiran and Litwin-Kumar [2] | Connectome-informed computational tools and connectome-constrained prediction.           | Separate structural source coverage, valid structure–function inference and the eligible artificial claim in the present portfolio.   | Not a competing toolbox or refutation of successful connectome-informed models.                                              |
| Hofmann et al., synaptic concept transfer [11]           | Implements biological synaptic-diversity concepts and evaluates artificial models.       | Trace why a source-inspired representation or learning label does not identify mechanism-specific utility in M2/M5.                   | No claim that biologically inspired transfer generally fails.                                                                |
| Muzellec and Kar, reverse predictivity [12]              | Distinguishes forward and reverse brain–model prediction.                                | Distinguish source biological evidence from artificial variable identity and task utility.                                            | Does not propose or validate a new alignment metric.                                                                         |
| Sun et al., dual memory pathways [13]                    | Prior fast–slow neuromorphic architecture and hardware co-design.                        | Tests a narrower mechanism $\times$ correspondence contrast, conditional task gate and fixed-epoch scaling in the present generators. | The abstraction is source-inspired, not a reproduction or new fast–slow architecture.                                        |
| Hupkes et al., GenBench taxonomy [15]                    | Operational dimensions for describing generalization studies and literature analysis.    | Use explicit case decisions to connect neural source claims, representations, effective controls and eligible conclusions.            | Categories and structured evaluation are not claimed as novel; no field-representative survey or validated taxonomy utility. |
| McGreivy and Hakim, weak baselines/reporting biases [16] | Systematic analysis of comparator and reporting practices in fluid-related PDE research. | Couple comparator checks to biological claim identity and control algebra in selected transfer cases.                                 | No estimate of bias prevalence in NeuroAI and no borrowing of the prior review's representativeness.                         |
| Wang et al., negative transfer [17]                      | Formal comparator-dependent source-task transfer failure and an avoidance method.        | Distinguish source-to-artificial abstraction from conventional source/target training transfer.                                       | A failed gate here is not automatically formal negative transfer or biological failure.                                      |

The proposed increment is the documented case-level change in inference and informative comparison. Whether that increment is sufficiently substantial for NMI remains an editorial and author judgment; the table does not establish priority or universal validity.

## **Supplementary Table S9 — How evidence decomposition changes the decision**

These retrospective decisions apply to the evaluated cases, not to all neural transfer studies. Source support identifies the observation and scope; computational interpretation identifies the proposed operation; correspondence identifies task-relevant relations; utility identifies performance under stated controls. Mechanism specificity remains a separate contrast. No score or causal ordering is assigned to these categories.

| Case                        | Tempting inference                                                                       | What the retained evidence changes                                                                                                               | Informative comparison or decision                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| M2 execution-state input    | Extra source-inspired information should improve control.                                | Signal availability and its biological identity do not determine utility in one learned controller. Representation/training explanations remain. | Compare shared-task controllers and information-recovery diagnostics; restrict the finding to the tested AcRKN and budget.                |
| M5 temporal correspondence  | A positive correspondence effect identifies the local learning rule.                     | The same representation supports larger generic effects; heads/objectives differ, so rule specificity is unidentified.                           | Retain the same-feature generic learner; require an actual matched rule contrast before claiming specificity.                             |
| DMP fast–slow state         | A matched dynamical contrast or small-data gain establishes mechanism-specific transfer. | Interpretation is conditional on task viability and interaction direction. Fixed epochs couple dataset size to update count.                     | Read each family gate separately; report fixed-epoch scaling without a pure-efficiency mechanism claim.                                   |
| Visual-cortex-inspired test | Nominal selective/global cells compare two contextual mechanisms.                        | Normalization makes the global control a constant offset. Passing a prevalence reference does not demonstrate contextual learning.               | Derive effective information and check metric/reference semantics before mechanism interpretation; preserve the frozen procedural record. |
| Fish1.5                     | An association test and a rewired null jointly settle structural mechanism validity.     | An unsupported positive association and an invalid topology null are distinct outcomes in one specimen.                                          | Keep the invalid null unestimated; do not treat neurons or null draws as independent animals.                                             |

**Supplementary Table S10 — Leave-one-case-out argument audit**

This is a logical dependence audit, not a rerun, meta-analysis or quantitative robustness test. The central bounded proposition is that the retained cases exhibit distinct breaks between source support, correspondence, specificity and utility, and that specifying the evidence question changes the eligible comparison. A case's deletion can leave that proposition supported while removing its particular empirical illustration. No deletion proves broad generality, representativeness or sufficiency of NMI novelty.

| Removed case                                    | Specific evidence no longer supplied                                               | Remaining support for the narrower synthesis                                                                     | Claim that must not survive deletion                                                                                                |
|-------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| M0 recovered calibration/falsification sequence | Bounded historical reservoir result under the M0 workstream label.                 | Fish1.5 and all four synthetic studies retain distinct validity, information, comparator and utility boundaries. | A completed M0 pilot or main-experiment claim; neither is documented as completed, and M0 is not necessary to the central argument. |
| Fish1.5                                         | Single-specimen association and invalid-topology-null example.                     | M2/M5/DMP and the visual test retain distinct artificial inference failures.                                     | Claims about observed structure–function association or topology-null validity from this specimen.                                  |
| Visual-cortex-inspired test                     | Control-algebra, AP-reference and visual interaction incident.                     | M2 separates information/utility; M5 separates correspondence/specificity; DMP adds conditional family gates.    | The specific normalized-control finding or contextual-learning interpretation.                                                      |
| M2                                              | Direct execution-state channel's controller result.                                | M5/DMP/visual tests still distinguish representation, comparator and utility.                                    | A result about added execution-state information.                                                                                   |
| M5                                              | Positive correspondence sensitivity with stronger same-feature generic prediction. | M2/DMP/visual tests still bound artificial utility and specificity.                                              | Evidence that this local rule uses temporal correspondence.                                                                         |
| DMP                                             | Three-family dynamical interactions and fixed-epoch scaling.                       | M2/M5/visual tests retain separate information, correspondence and effective-control decisions.                  | A three-family or fast–slow scaling claim.                                                                                          |

The source-supported worm and cerebellar computations remain literature-scoped motivation; deleting either source example removes that biological illustration without turning any synthetic result into biological evidence. Case selection is outcome-informed, and the 39-row disposition matrix counts records rather than independent replications.

## Remaining assembly work

Study-level procedures for the selected historical M0 sequence, Fish1.5, source-readout, M2 V1, M5 V1, DMP V1 and prospective V1 analyses are in the main Methods; all six main figures have source-data tables and export hashes. The technical interpretation and evidence traceability review is active. Remaining gates include exact source-to-claim crosswalk adjudication for some branch records, an owner-approved source snapshot, final citation/attribution review, reuse-rights audit, author-supplied end matter and a human novelty/narrative decision. The manuscript is an internal candidate and is not submission-ready.